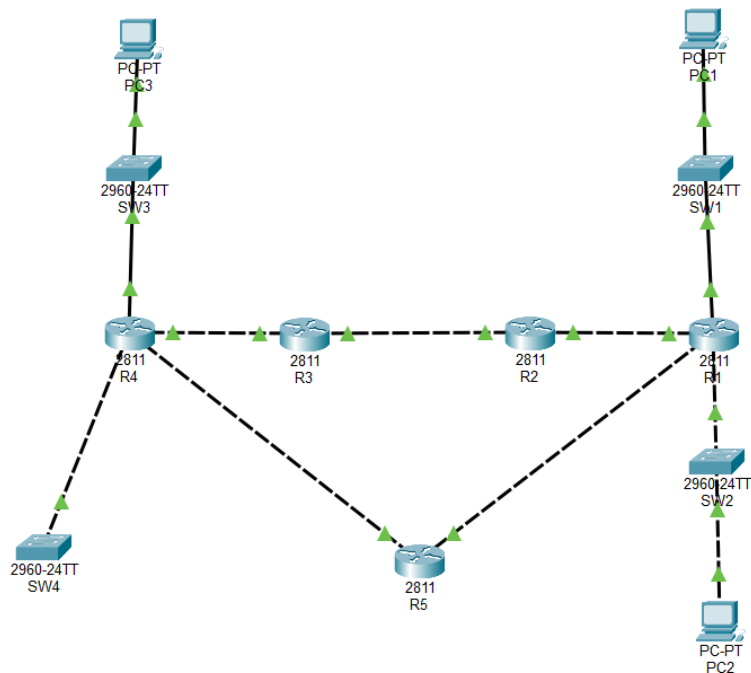


Week 8: Dynamic Routing Protocols (5 Marks)



74) R1:

```
Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Router(config-router)#network 10.0.0.0
Router(config-router)#no auto-summary
Router(config-router)#exit
Router(config)#
```

R2:

```
Router>ena
Router#router rip
      ^
% Invalid input detected at '^' marker.

Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Router(config-router)#network 10.0.0.0
Router(config-router)#no auto-summary
Router(config-router)#no shut
      ^
% Invalid input detected at '^' marker.

Router(config-router)#exit
Router(config)#
```

R3:

```
Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Router(config-router)#network 10.0.0.0
Router(config-router)#no auto-summary
Router(config-router)#exit
Router(config)#
```

R4:

```
Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Router(config-router)#network 10.0.0.0
Router(config-router)#no auto-summary
Router(config-router)#exit
Router(config)#
```

R5:

```
Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#network 10.0.0.0
^
% Invalid input detected at '^' marker.

Router(config)#router rip
Router(config-router)#network 10.0.0.0
Router(config-router)#no auto-summary
Router(config-router)#exit
Router(config)#
```

75)

```
Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#debug ip rip
^
% Invalid input detected at '^' marker.

Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#debug ip rip
RIP protocol debugging is on
Router#RIP: sending v1 update to 255.255.255.255 via FastEthernet0/0 (10.0.1.1)
RIP: build update entries
    network 10.0.2.0 metric 1
RIP: sending v1 update to 255.255.255.255 via FastEthernet0/1 (10.0.2.1)
RIP: build update entries
    network 10.0.1.0 metric 1
```

76) R1

```
Router>ena
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#exit
Router(config)#
```

R2:

```
Router>ena
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#exit
Router(config)#
```

R3:

```
Router>ena
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#exit
Router(config)#
```

R4:

```
Router>ena
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#exit
Router(config)#
```

R5:

```
Router>ena
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#exit
Router(config)#
```

77)

```
% Unknown command or computer name, or unable to find computer address

Router#RIP: sending v1 update to 255.255.255.255 via FastEthernet0/0 (10.0.1.1)
RIP: build update entries
      network 10.0.2.0 metric 1
RIP: sending v1 update to 255.255.255.255 via FastEthernet0/1 (10.0.2.1)
RIP: build update entries
      network 10.0.1.0 metric 1

Router#no debug all
All possible debugging has been turned off
Router#exit
```

80)

R1:

```
Router>
Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 10.0.0.0 0.255.255.255 area 0
Router(config-router)#exit
Router(config)#
```

R2:

```
Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 10.0.0.0 0.255.255.255 area 0
Router(config-router)#exit
Router(config)#
```

R3:

```
Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 10.0.0.0 0.255.255.255 area 0
Router(config-router)#exit
Router(config)#
```

R4:

```
Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 10.0.0.0 0.255.255.255 area 0
Router(config-router)#exit
Router(config)#
```

R5:

```
Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 10.0.0.0 0.255.255.255 area 0
Router(config-router)#exit
Router(config)#
```

81)

```
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.0.1.0/24 is directly connected, FastEthernet0/0
L       10.0.1.1/32 is directly connected, FastEthernet0/0
C       10.0.2.0/24 is directly connected, FastEthernet0/1
L       10.0.2.1/32 is directly connected, FastEthernet0/1
Router#
```

82)

```
Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface fastethernet 0/0
Router(config-if)#shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to down

Router(config-if)#exit
Router(config)#
```

IP ROUTE:

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.2.0/24 is directly connected, FastEthernet0/1
L       10.0.2.1/32 is directly connected, FastEthernet0/1
Router#
```

IP OSPF DATABASE:

```
Router#show ip ospf database
        OSPF Router with ID (10.0.2.1) (Process ID 1)

                Router Link States (Area 0)

Link ID      ADV Router   Age         Seq#         Checksum Link count
10.0.2.1     10.0.2.1     204         0x80000003  0x003eed 1
Router#
```

83)

```
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface fastethernet 0/0
Router(config-if)#shutdown
Router(config-if)#exit
Router(config)#
```

```
Link ID          ADV Router      Age           Seq#           Checksum Link count
10.0.2.1         10.0.2.1        204          0x80000003    0x003eed 1
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.2.0/24 is directly connected, FastEthernet0/1
L       10.0.2.1/32 is directly connected, FastEthernet0/1
Router#
```

84) Traffic Type:

- RIPv1: Uses broadcast 255.255.255.255. Every device on the LAN hears it, even if they don't run RIP.
- RIPv2: Uses multicast 224.0.0.9. Only routers that joined the RIPv2 multicast group process it. More efficient.

85) Because RIPv2 does equal-cost load balancing by default (up to 4 paths). If there are two paths with the same hop count/metric to 10.1.1.0/24, RIP will install both in the routing table.

86) No, OSPF has an Administrative Distance of 110, RIP has 120.

87)